

ASHAR RAIS

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RESEARCH PROFILE

Post-graduate in cellular and molecular biology with research experience in hematopoietic stem cell biology, blood disorders and mechanobiology.

EDUCATION

- **Master of Science (M.Sc.) in Bioscience (Cell & Molecular Biology specialisation) (2023 – 2025)**
Bioscience department, King Abdullah University of Science & Technology (KAUST), Kingdom of Saudi Arabia (KSA)
CGPA: **3.89/4**
 - **Bachelor of Science (Hons.) in Biochemistry (2021 – 2023)**
Dept. Of Biochemistry, Aligarh Muslim University, India
CGPA: **8.6/10**
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RESEARCH EXPERIENCE

Master's Thesis Research (Aug 2024 – May 2025)

Cell Migration & Signalling Lab, KAUST

Project: *Favism's Overlooked Gene: Examining How a Mutation in CD34 Determines the Severity of G6PD Deficiency*

Supervisor: Prof. Jasmeen Merzaban

Description: A collaborative project between the Cell Migration & Signalling lab and Dr. Naif Almontashiri's lab at Taibah University. This study aims to explore consequences of a CD34 (classical marker of hematopoietic stem cells) mutation upon the functioning of HSCs and how this might affect disease progression in a G6PD patient. With no data till date on the pathogenicity of CD34, this study is the first of its kind and holds great potential in opening new avenues to further understand well-studied blood disorders.

- Used RT-qPCR and multiple proteomics-based techniques to characterise the expression of CD34 & mutant G6PD introduced into HL-60 using lentiviral transduction.
- Studied protein-protein interactions using immunoprecipitation and molecular dynamics simulations.
- Evaluated the effect of CD34 introduction upon HL-60 cell rolling via microfluidics-based apparatus.

Master's Research Project-II (Feb 2024 – Jul 2024)

Cell Migration & Signalling Lab, KAUST

Project: *Exploring the Mechanobiological Role of LINC complex in Stem Cell Homing Paradigm*

Mentor: Dr. Mansour Aldehaiman

- Studied the expression levels of LINC complex proteins (SUN proteins and Nesprins) in different human cell lines using Immunoblotting.
- Used confocal microscopy to visualise the sub-cellular location of LINC complex proteins.
- Prepared culture media of different viscosity using methylcellulose and used viscometer to analyse media viscosity.

Master's Research Project-I (Oct 2023 – Nov 2023)

Vibrational Imaging Lab (VibraLab), KAUST

Project: *Lipid Droplets: Their Role in Cancer Stem Cell Survival under Hypoxic Conditions*

Mentor: Dr. Alperen Serdaroglu

- Studied the effects of hypoxic conditions upon the lipid profile of CC09 cells.

PUBLICATIONS

- **Rais A**, Husain A, Hasan GM, Hassan MI; A review on regulation of cell cycle by extracellular matrix; International Journal of Biological Macromolecules, Volume 232, 2023, 123426, ISSN 0141-8130.
<https://doi.org/10.1016/j.ijbiomac.2023.123426>.
- Nasif Hussain, Mohd Moin Khan[#], **Ashar Rais[#]**, Nawaz Akhter, Mudassir Meraj Banday, Mohd Mohsin, Asifa Khan and Rizwan Hasan Khan [[#] These authors contributed equally to this work]; Rewriting the Neurodegenerative Code: How Non-coding RNAs Drive Epigenetic and Inflammatory Mechanisms. (*Under Review*)

TECHNICAL SKILLS

Cell culturing

Human cell lines (HEK-293, MDA-MB-231, HL-60 and KG-1a)

Proteomics

Immunoblotting, Immunoprecipitation (IP), co-IP, Flow Cytometry, Mass Spectrometry

Transcriptomics

RT-qPCR and Lentiviral Transduction.

Imaging

Confocal Microscopy and Axio-scan Imaging

Assays

Glycosylation Profile Assays, Microfluidics-based Cell Migration Assay, Cell Cycle Analysis using PI-staining, Biochemical Assays for Metabolite Detection

Computational skills

Agent-based Model (ABM) designing in Python and Molecular Dynamic Simulations.

COMPETITIVE FELLOWSHIPS & NATIONAL EXAMINATIONS

- GATE-XL (2026): Qualified with a National Rank of 927.
- KAUST Fellowship (2023)

LEADERSHIP, ACADEMIC & ENTREPRENEURIAL EXPERIENCE

- **Co-Founder – HEMSCAN Diagnostics LLP (2025–Present)**

Developing a point-of-care diagnostic platform for rapid blood biomarker screening.

REFERENCES

- **Prof. Jasmeen Merzaban** | MS Thesis Supervisor | jasmeen.merzaban@kaust.edu.sa
Bioscience department, King Abdullah University of Science & Technology (KAUST), Kingdom of Saudi Arabia (KSA)
- **Prof. Christian Froekjaer Jensen** | cfjensen@kaust.edu.sa
Bioscience department, King Abdullah University of Science & Technology (KAUST), Kingdom of Saudi Arabia (KSA)
- **Prof. Samah Zeineb Gadhoun** | Samah.GADHOUM@kaust.edu.sa
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